Programming Fundamentals Lab

Computer Science



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Task-1

Write a program to print positive number entered by the user. If user enters negative number print number entered is positive otherwise print number is negative.

```
#include <iostream>
using namespace std;
int main ()
{

 double num;
cout<< "Enter a number: ";
 cin >> num;

if (num>= 0) {
 cout << "Number entered is positive"<<
endl;
} else {
 cout << "Number entered is negative"<<
 endl;
 }
return 0;
}
```

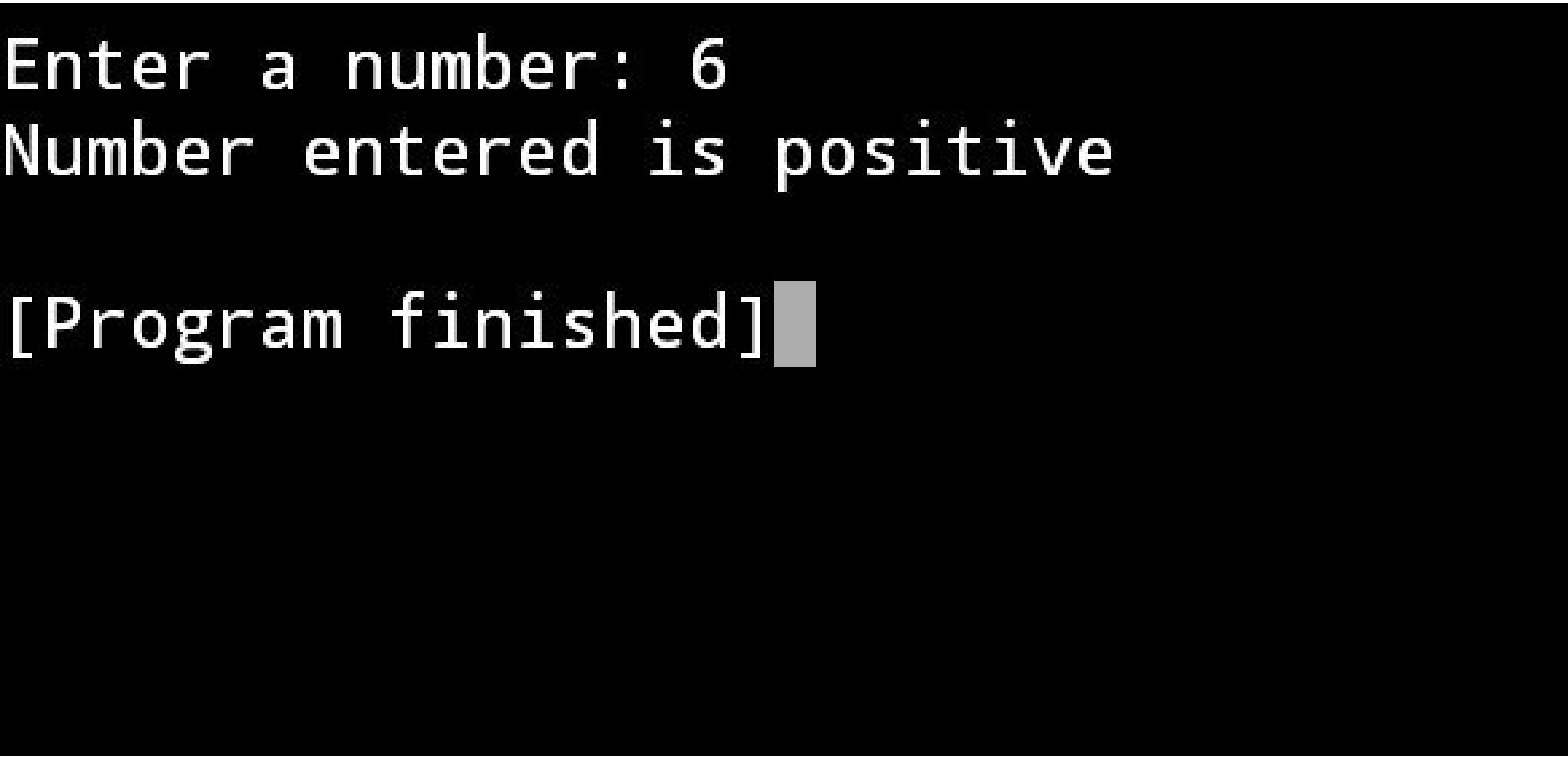
Output:

```
Enter a number: 6
Number entered is positive
```

```cpp
#include <iostream>
using namespace std;
int main()
 {
    double num;

    cout << "Enter a number: ";
    cin >> num;
     if (num>=0)
    {
        cout << "Number entered is positive" << endl;
    }
    else
    {
        cout << "Number entered is negative" << endl;
    }

    return 0;
}
```


Enter a number: 6
Number entered is positive
[Program finished]

Task-2

Program to check whether an integer is positive, negative or zero.

```
#include <iostream>
using namespace std;
int main()
{
   int num;
  cout << "Enter a number";
  cin >> num{
  if(num >0);
   cout << "The number is positive" << endl;
   }
   else if (num < 0){
    cout << "The number is negative" << endl;
  }
   else {
     cout << "The number is zero" << endl;
  }
   return 0;
  }
```

Output:

Enter an integer: 55

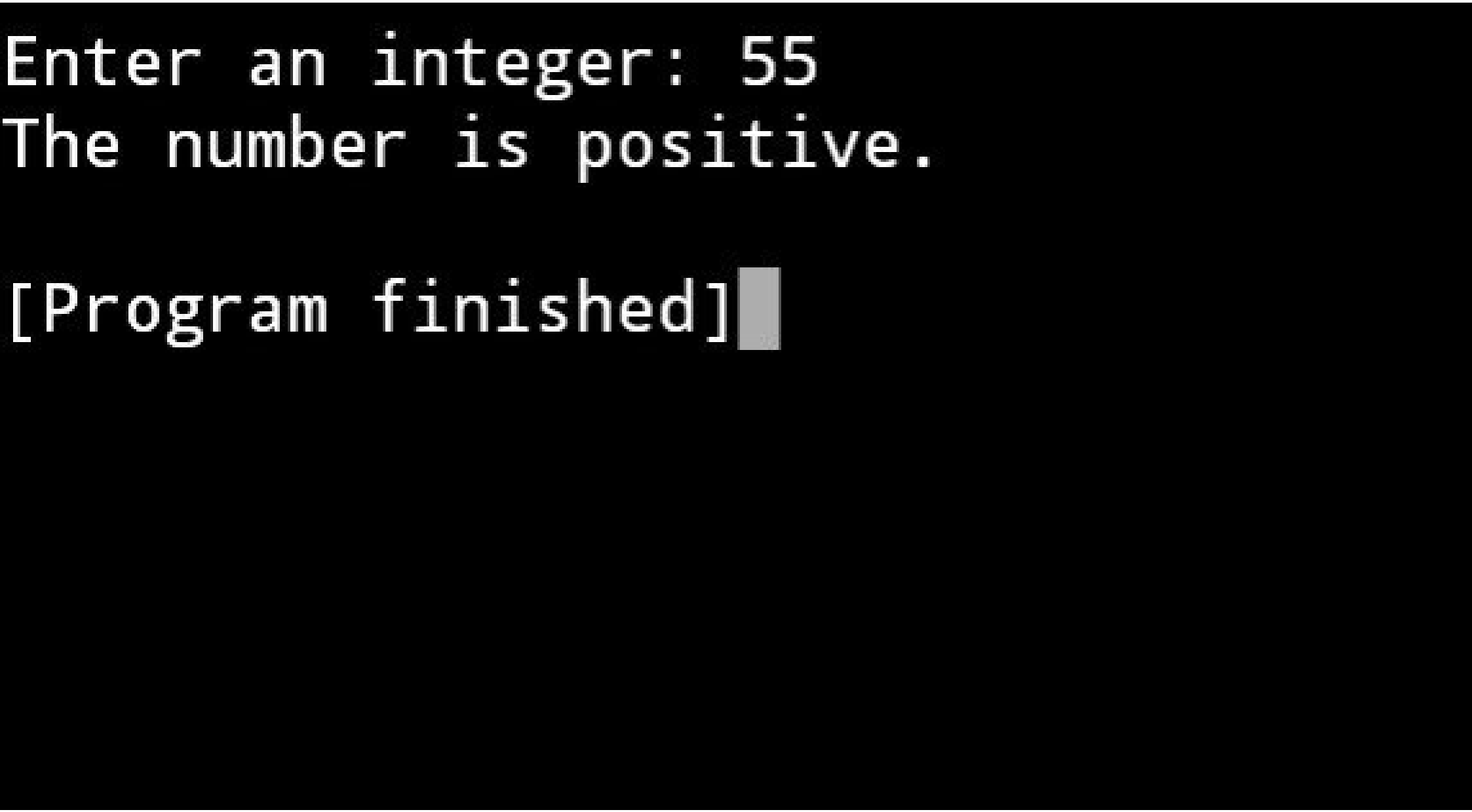
The number is positive

```cpp
#include <iostream>

using namespace std;

int main() {
    int num;
    cout << "Enter an integer: ";
    cin >> num;
    if (num > 0) {
        cout << "The number is positive." << endl;
    }
    else if (num < 0) {
        cout << "The number is negative." << endl;
    }
    else {
        cout << "The number is zero." << endl;
    }

    return 0;
}
```


Enter an integer: 55
The number is positive.
[Program finished]

Task-3

Input : Mark

Process: If mark greater than and equal to 75, score will be A

If mark less than 75 and greater than and equal to 60, score will be B

If mark less than 60 and greater than and equal to 45, score will be C

If mark less than 30, score will be D

Output: Print the grade of your score.

```
#include <iostream>
using namespace std;
int main ()
{
    int marks;
    cout << "Enter your marks:";
    cin >> marks;
    if (marks >= 75) {
      cout << "Score will be A" << endl;
      }
      else-if(marks < 75 && marks >= 60) {
cout << "score will be B" << endl;
  }
    else if (marks < 60 && mask >= 45) {
cout << "score will be C" << endl;
  }
    else if (marks < 30) {
cout << "score will be D" << endl;
```

```
    }
    return 0;
}
```

Output:

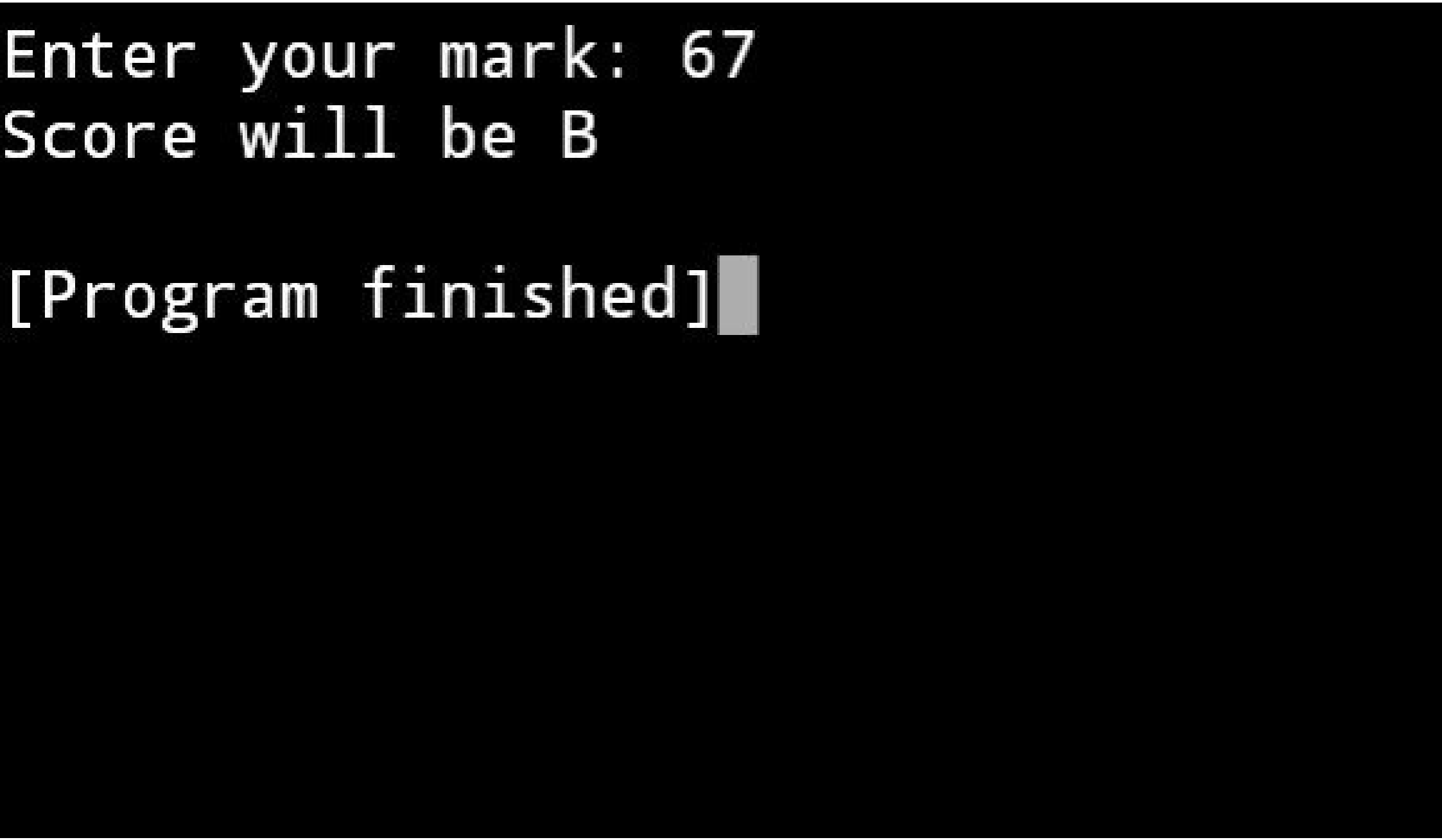
```
Enter your mark : 67
Score will be B
```

```cpp
#include <iostream>

using namespace std;

int main() {
    int marks;
    cout << "Enter your mark: ";
    cin >> marks;
    if (marks >= 75) {
        cout << "Score will be A" << endl;
    }
    else if (marks < 75 && marks >= 60) {
        cout << "Score will be B" << endl;
    }
    else if (marks < 60 && marks >= 45) {
        cout << "Score will be C" << endl;
    }
    else if (marks < 30) {
        cout << "Score will be D" << endl;
    }

    return 0;
}
```


Enter your mark: 67
Score will be B
[Program finished]

Task-4

Find Largest Number using Nested if.... statement. Check whether the number entered by the user is positive or not. If it is positive then calculated how many digits the number have.

```
# include <iostream>
using namespace std;
 int main ()
 {

    int n1, n2, n3;
  cout<< "Enter three numbers: ";
   cin >> n1>>n2>> n3;
      int Largest;
   if (n1>= n2) {
         if (n1>= n3) {
      Largest = n1;
   } else {
      Large = n3;
```

```
    }
  } else {
      if (n2 >= n3) {
      largest = n2;
    }else {
        largest = n3;
      }
    }
cout<<"Largest Number is:"<< large << endl;
  int num;
cout <<"\n Enter a number L check digits:";
     cin >> num;
  if(num > 0) {
   cout <<" Number is Positive" << endl;
     int cout = 0
     int temp = num;
    while (tem / = 0) {
       temp = temp /10;
       cout ++;
    }
    cout <<" Number of digits:" << cout << endl;
  }else {
   cout <<" Number is not positive" << endl;
  }
```

```
    return 0;
}
```

Output:

```
Enter three numerbers: 6 9 3
Largest Number is: 9
```

```cpp
#include <iostream>

using namespace std;

int main() {
    int n1, n2, n3;
    cout << "Enter Three numbers: ";
    cin >> n1 >> n2 >> n3;

    int largest;
    if (n1 >= n2) {
        if (n1 >= n3) {
            largest = n1;
        } else {
            largest = n3;
        }
    } else {
        if (n2 >= n3) {
            largest = n2;
        } else {
            largest = n3;
        }
    }

    cout << "Largest Number is: " << largest << endl;

    int num;
    cout << "\nEnter a number to check digits: ";
    cin >> num;

```

```
    if (num > 0) {
        cout << "Number is Positive" << endl;
        int count = 0;
        int temp = num;
        
        while (temp != 0) {
            temp = temp / 10;
            count++;
        }

        cout << "Number of digits: " << count << endl;
    } else {
        cout << "Number is not positive" << endl;
    }

    return 0;
}
```